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Prepared by

Mohamed Raed BOUALI

Intelligent Multilingual OCR for Scanned Documents Using Vision-Language Models

Professional Supervisors: **EYA BOUHOUCH**

MOHAMED ANIS AOUADI

EY Tunisia

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I authorize the student to submit their internship report for a defense.

Professional Supervisors, **EYA BOUHOUCH** & **MOHAMED ANIS AOUADI**

Signature and stamp

Dedication

This report is dedicated to all those who value rigor, perseverance and excellence in the field of artificial intelligence and document intelligence.

May it stand as a testament to the importance of methodical work and innovation in solving the complex challenges faced by modern organizations.

Through this project, I affirm my commitment to actively contributing to the development of effective, trustworthy solutions tailored to the contemporary challenges of intelligent document processing.

Acknowledgments

I would like to express my deep gratitude to everyone who contributed to the completion of this project.

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I also thank my colleagues and the EY Tunisia team for their warm welcome, their collaboration and the enriching exchanges that supported my learning.

Finally, I would like to thank my family for their unwavering moral support, which encouraged me to persevere through this important stage of my professional journey.

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List of Abbreviations

- **AI** = Artificial Intelligence
- **API** = Application Programming Interface
- **CBS** = Core Business Services
- **CER** = Character Error Rate
- **DPI** = Dots Per Inch
- **GPU** = Graphics Processing Unit
- **JWT** = JSON Web Token
- **LLM** = Large Language Model
- **OCR** = Optical Character Recognition
- **PDF** = Portable Document Format
- **REST** = Representational State Transfer
- **RTL** = Right-To-Left (writing direction)
- **SPA** = Single-Page Application
- **SaT** = Strategy and Transactions
- **VLM** = Vision-Language Model
- **VRAM** = Video Random-Access Memory
- **WER** = Word Error Rate

Chapter 1

Presentation of the Host Organization

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1.1 EY: Who We Are

EY (Ernst & Young) is one of the world leaders in **audit**, **consulting**, **tax and law**, and **strategy and transactions**. Across the globe, EY's expertise and the quality of its services help build trust in the economy and in financial markets. The firm develops talent so that, together, its people support organizations on the path to sustainable growth. In doing so, EY plays an active role in building a fairer and more balanced world for its teams, its clients and society as a whole.



Figure 1.1: EY logo

1.2 Key Figures Worldwide

EY's global footprint reflects its scale as well as the consistency of its standards across borders, as Table 1.1 shows.

Table 1.1: EY worldwide in key figures

Indicator	**Value**
Professionals	400,000
Offices	750
Countries	152
Unwavering commitment	1

EY's people describe themselves as professionals of integrity, showing mutual respect and team spirit; dynamic teams who share the enthusiasm and courage of leaders; and men and women who know how to build fair and lasting relationships.

1.3 EY Tunisia

With a local presence since **1987** (more than 30 years), **AMC Ernst & Young** is one of the leading professional-services firms in Tunisia, thanks to more than **900** professionals permanently based in its

Tunis office and organized across several service lines: **Assurance**, **Consulting**, **Tax**, **Strategy & Transactions**, and **Core Business Services (CBS)**.

Table 1.2: EY Tunisia in key figures

Indicator	**Value**
Collaborators	+900
Partners	23
Sectors of activity	10
Main service lines	4

The EY Tunisia office has supported companies and public organizations for more than 30 years, carrying out over **600 engagements in 30 countries** each year. In terms of local competitiveness, EY Tunisia is the **1st business-consulting firm in Tunisia** and hosts the **1st EY Cyber & RPA lab in Tunisia**.

1.4 Service Lines and Careers

EY Tunisia brings together a wide range of competencies, and therefore many career perspectives. Table 1.3 details the competencies covered by each service line; the **AI and Data** team, within which this internship took place, belongs to the **Consulting** line.

Table 1.3: Service lines of EY Tunisia and their main competencies

Service line	**Main competencies**
Assurance	Audit; Financial Accounting Advisory Services & Risk; Climate Change and Sustainability Services; Forensic & Integrity Services; Managed Services; Technology Risk.
Consulting	Business Transformation; Supply Chain & Operations; Financial Services Transformation; Actuarial Services; People Consulting; Innovation & Experience Design; Technology Strategy & Transformation; **AI and Data**; Digital Engineering; Platforms (Microsoft); Cyber Security.
Strategy & Transactions	Transaction Diligence; Valuation, Modeling & Economics; Lead Advisory; Corporate and Growth Strategy; Turnaround and Restructuring Strategy; Transaction Strategy and Execution.
Tax	Global Compliance and Reporting; Business Tax Services and Advisory; International Tax Advisory; Transaction Tax Services; People Advisory Services; Entity Compliance and Governance Services; Labor & Employment Law advice.

1.5 Core Business Services (CBS)

Alongside the client-facing service lines, the **Core Business Services (CBS)** teams provide the internal support that keeps the firm running. Their respective responsibilities are given in Table 1.4.

Table 1.4: Core Business Services teams and their responsibilities

Team	**Main responsibilities**
Talent Team	Recruitment; onboarding follow-up; personnel administration and payroll; internship management; the *EY Academy* for training and skills development; career management; internal communication and well-being.
IT Service	IT support.
Finance and Control	Cost accounting and billing.
Facilities	Travel; buildings; hospitality; procurement and general services.
Markets & Business Development (MBD)	Marketing projects; digital communication; reporting; bid support.
Risk Management	Management of business, client and engagement risks.

1.6 Organization of EY Tunisia

At the top of the local structure, the **Country Managing Partner (CMP)** oversees the firm's activities, supported by the leaders of the core service lines: the **Assurance Leader**, the **Consulting Leader**, the **Tax Leader** and the **Strategy & Transactions (SaT) Leader**. My internship took place within the **Consulting** service line, in the **AI and Data** team, which delivers applied artificial-intelligence and data-engineering solutions to EY's clients – the natural home for an intelligent document-processing project such as the one presented in this report.

Chapter 2

Project Context and State of the Art

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2 Classical OCR and Its Limits 7

3 Vision-Language Models 7

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2.1 The Document Digitization Problem

Organizations produce and receive large volumes of **scanned documents**: contracts, invoices, administrative forms, identity documents, and official circulars. Stored as images, these documents cannot be searched, indexed, aggregated or fed into downstream processes. **Optical Character Recognition (OCR)** is the technology that converts such images back into machine-readable text, and it is the foundation of any intelligent document-processing workflow.

In practice, real-world scans are far from ideal. They are often **noisy, skewed, low-contrast or of low resolution**; they mix **printed and handwritten** content; they contain **complex layouts** such as tables, multi-column text and checkboxes; and, in the Tunisian context, they frequently combine **French, English and Arabic**, the latter being written **right-to-left**. These characteristics make robust recognition a genuinely difficult problem.

2.2 Classical OCR and Its Limits

Classical OCR engines – the best known being the open-source **Tesseract** – detect the shapes of individual characters and assemble them into words using language models trained per script. They are fast, lightweight and effective on clean, printed, single-language documents.

However, classical OCR has well-known limitations:

- It degrades sharply on **noisy, faint or distorted** scans, and on **handwriting**.
- It has **no understanding of layout**: reading order, tables and form structure are typically lost, producing a flat stream of text.
- It handles **mixed scripts** poorly, and Arabic in particular – with its cursive joining and right-to-left direction – remains challenging.
- It does not capture **semantic elements** such as which checkbox in a form is ticked.

2.3 Vision-Language Models

Recent progress in deep learning has produced **Vision-Language Models (VLMs)**: neural networks that take an image *and* a textual instruction as input and generate text as output, within a single model. Unlike classical OCR, a VLM **understands** the content it reads – context, layout, language and structure – which allows it to:

- transcribe degraded scans and handwriting far more robustly;

- preserve document structure, emitting headings, lists and tables (for example as Markdown);
- handle multilingual and right-to-left text natively;
- interpret semantic cues such as checkbox states.

Several open VLMs are now available and can run on commodity hardware. Among them, the **Qwen3-VL** family stands out for its strong multilingual coverage – including Arabic – and its document-oriented capabilities, which makes it a natural choice for this project.

Table 2.1 summarizes the comparison between the two families of approaches on the criteria that matter for this project.

Table 2.1: Classical OCR versus Vision-Language Models

Criterion	**Classical OCR (Tesseract)**	**Vision-Language Model**
Clean printed text	Excellent	Excellent
Noisy, faint or skewed scans	Degrades sharply	Robust
Handwriting	Very limited	Handled
Layout understanding	None; flat text stream, reading order lost	Preserved; headings, lists and tables emitted as Markdown
Mixed scripts and Arabic (right-to-left)	Poor	Native
Semantic elements (e.g. checkbox states)	Not captured	Interpreted
Resource footprint	Light, CPU-only	Requires a GPU

2.4 Local versus Cloud Inference

VLMs can be consumed in two ways. **Cloud APIs** (such as Groq) offer very large models with no local hardware requirement, but every document must be transmitted to a third party – unacceptable for confidential material – and usage is metered and dependent on network availability and the provider's catalogue. **Local inference**, by contrast, runs the model on the organization's own hardware: no data leaves the machine, there is no per-page cost, and the system works fully offline. The trade-off is that the model size is bounded by the available **GPU memory (VRAM)**.

Table 2.2 contrasts the two deployment modes.

Table 2.2: Local inference versus cloud API

Criterion	**Local inference**	**Cloud API (e.g. Groq)**
Data confidentiality	Total: no document leaves the machine	Every document is transmitted to a third party
Cost	No per-page cost	Metered usage
Network dependency	Works fully offline	Requires availability of the network and of the provider
Model size	Bounded by the available GPU memory (VRAM)	Very large models available
Hardware requirement	A local GPU is needed	None
Model catalogue	Limited to what fits locally	Dependent on the provider's catalogue

For a firm such as EY, which handles highly confidential client documents, the **local, offline** option is decisive from a data-governance standpoint. This project therefore adopts a **local-first** design, using a cloud engine only as an optional benchmark.

2.5 Problem Statement and Objectives

Based on this context, the project set out to build an intelligent OCR system meeting the following objectives:

- accurately transcribe scanned documents in **French, English and Arabic**, including mixed-language and right-to-left content;
- preserve **document structure** (headings, tables, checkbox states);
- run the recognition engine **locally and offline** to guarantee confidentiality;
- expose the solution through a usable **web platform** with authentication and a per-user document history;
- provide a way to **measure and verify** recognition quality rather than assume it.

Chapter 3

Assignments and Work Carried Out

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1 **Study of OCR Approaches and Model Selection** 11

2 **Design of the Recognition Pipeline** 12

3 **Development of the Full-Stack Platform** 12

4 **Evaluation and Quality Assurance** 12

During my internship within the AI and Data team of EY Tunisia, I was entrusted with the design and construction of an intelligent, multilingual OCR system. The work was organized around the following areas, which together cover the full lifecycle of an applied AI project. Table 3.1 gives an overview of these assignments and of the deliverable associated with each of them.

Table 3.1: Overview of the assignments and their deliverables

Assignment	**Main activities**	**Deliverable**
Study of OCR approaches and model selection	Comparison of classical OCR and VLMs; evaluation of candidate open models; assessment of the local hardware constraint	Choice of Qwen3-VL 4B-instruct served through Ollama
Design of the recognition pipeline	Image preprocessing; prompt strategy; per-page reading of PDFs; language detection and right-to-left handling	End-to-end recognition pipeline
Development of the full-stack platform	REST API; single-page application; authentication and per-user history	Web platform (FastAPI, React, MongoDB)
Evaluation and quality assurance	CER/WER metrics; tuning of inference parameters; two-model verification	Evaluation tooling and completeness-verification tool

3.1 Study of OCR Approaches and Model Selection

The first assignment consisted of surveying the available OCR approaches and selecting the most suitable engine for the target use case:

- Comparing **classical OCR** (Tesseract) with **Vision-Language Models** on degraded, multilingual and handwritten scans.
- Evaluating candidate open VLMs and their ability to handle French, English and Arabic.
- Assessing the constraint of running the model **locally** on the available hardware (a laptop GPU with 6 GB of memory).

This study led to the selection of **Qwen3-VL** in its 4-billion-parameter *instruct* variant, served locally through **Ollama**.

3.2 Design of the Recognition Pipeline

A second assignment was to design an end-to-end recognition pipeline rather than a single model call:

- An **image-preprocessing** stage (grayscale, denoising, deskewing, contrast enhancement) to improve legibility.
- A **prompt strategy** instructing the VLM to transcribe every element faithfully – including small print, tables, and checkbox states – without translating or summarizing.
- A **per-page reading strategy** for PDF documents, which are rasterized locally page by page.
- Automatic **language detection** and right-to-left handling for Arabic passages.

3.3 Development of the Full-Stack Platform

The recognition core was exposed through a complete web platform, whose development represented the bulk of the technical work:

- A **REST API** (FastAPI) orchestrating preprocessing, recognition and persistence.
- A **React** single-page application providing upload, engine selection, language hints, and a searchable document history.
- Authentication and a per-user history stored in **MongoDB**.

3.4 Evaluation and Quality Assurance

Finally, I was responsible for measuring and improving recognition quality:

- Implementing **Character Error Rate (CER)** and **Word Error Rate (WER)** metrics.
- Tuning inference parameters (context window, rasterization resolution) for the best speed/completeness trade-off on the target GPU.
- Designing a **two-model verification** tool that flags text on which two local models disagree, so that potential omissions can be reviewed.

Together, these assignments allowed me to take a real AI product from problem analysis to a working, evaluated system.

Chapter 4

Development and Implementation

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4.1 Overall Presentation of the Solution

The delivered system is a full-stack **document-intelligence platform** whose purpose is to transcribe scanned documents (images or PDFs) in French, English and Arabic into structured, machine-readable text. Its distinguishing feature is that the recognition engine runs a **Vision-Language Model locally**, so that confidential documents are never sent to a third party. The same recognition core is shared by a web application, a command-line interface and a lightweight prototype UI.

4.2 Global Architecture

The platform follows a classic three-tier architecture, with an interface-agnostic OCR core reused by every front end:

- **Front end**: a React (Vite) single-page application communicating with the back end through a REST API.
- **Back end**: a FastAPI service that handles authentication, orchestrates the OCR pipeline, and persists results.
- **Data store**: a MongoDB database holding users and their document history.
- **OCR core**: a Python package (`src/`) containing preprocessing, the recognition engines, language detection and evaluation, called identically by the back end, the CLI and the prototype.

4.3 Technologies Used

Figure 4.1 gathers the main building blocks of the platform, from the recognition layer down to the data store. The complete technology stack, with the role of each component, is summarized in Table 4.1.

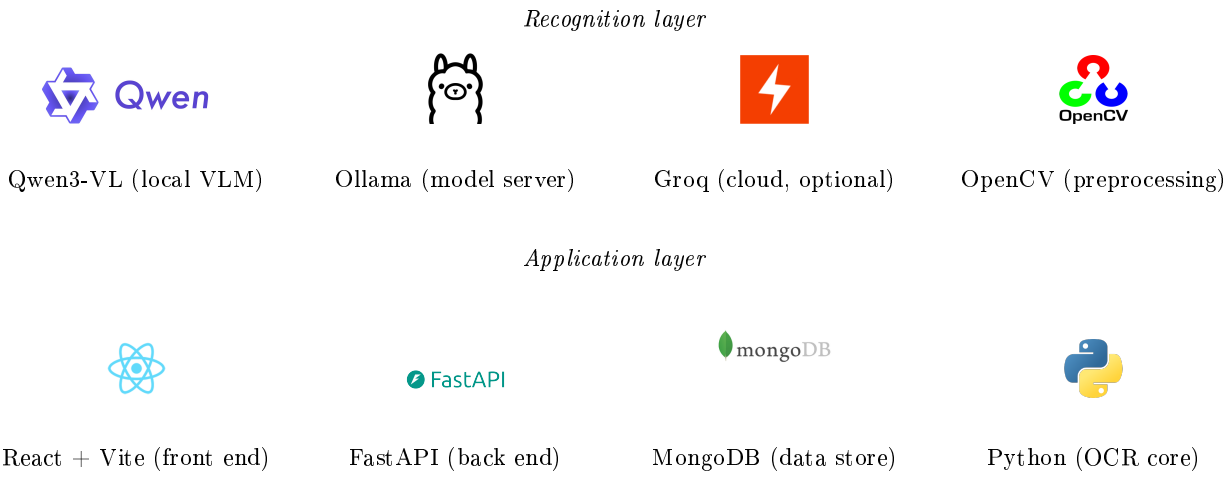


Figure 4.1: Technologies making up the platform, by layer

Table 4.1: Technologies used and their role in the platform

Technology	**Layer**	**Role in the platform**
Qwen3-VL (4B-instruct)	Recognition	Vision-Language Model performing the actual transcription; handles French, English and Arabic and preserves layout as Markdown.
Ollama	Recognition	Local model server running the VLM on the machine's GPU and exposing it through a simple API; makes the whole system fully offline.
FastAPI + Uvicorn	Back end	Python web framework and server providing the REST API, with JWT-based authentication.
React + Vite	Front end	Library and build tool for the single-page application.
MongoDB (PyMongo)	Data store	Document-oriented database storing users and extraction history.
OpenCV, Pillow	Preprocessing	Grayscale conversion, denoising, deskewing and contrast enhancement.
PyMuPDF	Preprocessing	Local rasterization of PDF pages into images, with no external dependency.
Groq API (optional)	Recognition	Cloud VLM engine (Llama 4 vision) kept as an interchangeable benchmark.

4.4 Processing Flow

When a document is submitted, the pipeline proceeds as follows:

- The file is validated and, if it is a PDF, **rasterized page by page** locally with PyMuPDF at 300 DPI.
- Each page image is **preprocessed** (grayscale, denoise, deskew, contrast) to improve legibility.
- The image is sent to the selected **VLM engine** (local by default) with an OCR-specific prompt instructing a faithful, structure-preserving transcription.
- The result undergoes **language detection**; Arabic-heavy text is rendered right-to-left.

- The extraction, together with its metadata (engine, languages, duration), is **stored** in MongoDB and returned to the interface, where it can be consulted, copied or exported as a PDF.

4.5 Working Environment

The system was developed and evaluated on the hardware and software environment described in Table 4.2.

Table 4.2: Hardware and software working environment

Element	**Specification**
Machine	Lenovo laptop (model 83DV)
Processor (CPU)	13th Gen Intel® Core™ i7-13650HX, 14 cores / 20 threads, 2.6 GHz base frequency
Graphics card (GPU)	NVIDIA GeForce RTX 4050 Laptop GPU, 6 GB of dedicated memory (VRAM)
Memory (RAM)	32 GB
Operating system	Windows 11 Enterprise (build 26200)
Language and runtime	Python 3.14, Node.js 22.18 (npm 10.9)
Model server	Ollama 0.32.3
Database	MongoDB 8.2.5
Development tools	Visual Studio Code 1.130, Git 2.50

The development tools used throughout the internship are shown in Figure 4.2.



Visual Studio Code (editor)



Git (version control)



Python (OCR core, CLI)

Figure 4.2: Development tools used during the internship

4.6 Performance Tuning

Running a VLM on a 6 GB laptop GPU required careful tuning. The most significant parameter proved to be the model's **context window**: an oversized value forced part of the model onto the CPU and tripled the time per page. Reducing it to a value sized for a single page kept the model fully GPU-resident and brought the processing time down to roughly **30 seconds per dense page**, with no loss of completeness. The PDF rasterization resolution (300 DPI) was likewise chosen to keep fine print legible while limiting cost. Table 4.3 reports the effect of this parameter.

Table 4.3: Effect of the context window on the placement and the speed of the model

Context window	**Placement of the model**	**Time per dense page**	**Completeness**
Default value (oversized)	Partially offloaded to the CPU	≈ 90 s	Unchanged
8192 tokens (sized for one page)	Fully resident on the GPU	≈ 30 s	Unchanged

Chapter 5

Case Study: Evaluation and Completeness Verification

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5.1 Project Context

A recurring requirement expressed during the internship was that the system should **miss as little text as possible** on sensitive documents – for example a confidential administrative form. Because no OCR system, local or cloud, can *guarantee* a zero error rate, the honest engineering answer is to **measure** recognition quality and to **surface** the places where text may have been missed, so that a human can review them. This case study describes how the system was evaluated and how a completeness-verification mechanism was designed.

5.2 Objectives

- Quantify recognition quality with standard, reproducible metrics.
- Compare the local engine against a cloud engine and against classical OCR.
- Provide a practical way to detect potential omissions on a given document.

5.3 Evaluation Methodology

Recognition quality was measured with two standard metrics implemented in the OCR core:

- **Character Error Rate (CER)**: the proportion of character-level edits (insertions, deletions, substitutions) needed to turn the output into the reference text.
- **Word Error Rate (WER)**: the same measure at the word level.

Both use Unicode normalization, which is essential for consistent comparison of Arabic text. A reference (ground-truth) transcription is compared against the system output to obtain the scores.

5.4 Completeness Verification

To address the "do not miss anything" requirement, a **two-model verification** tool was developed. The same page is transcribed by **two independent local models** (Qwen3-VL 4B and 8B), and their outputs are aligned word by word:

- where the two models **agree**, confidence is high;
- where they **disagree**, the span is flagged for human review, marking explicitly when one model produced text the other omitted.

The rationale is that two models have **different blind spots**: in testing, the smaller model was stronger on some fine print and checkboxes, while the larger one recovered headers the smaller one

dropped. Diffing them therefore reliably surfaces real omissions that a single pass would hide. This turns completeness from an unverifiable claim into a reviewable, measurable property.

5.5 Results Obtained

Table 5.1 reports the measurements obtained on the reference documents, and Table 5.2 compares the two local models on the target hardware.

The evaluation criterion retained for the reference run was **completeness** – the share of the expected content actually transcribed, scored out of ten by manual review – together with the processing time. The CER and WER metrics described in Section 5.3 are implemented in the OCR core and are intended for the larger annotated corpus identified as future work in Section 5.6.

Table 5.1: Measurements obtained on the reference documents

Engine	**Document type**	**Completeness**	**Time / page**
Qwen3-VL 4B (local)	Dense handwritten form	10/10	59 s
Qwen3-VL 8B (local)	Dense handwritten form	9/10	135 s
Qwen3-VL 4B (local)	Dense page, after context tuning	Unchanged	≈ 30 s
Qwen3-VL 4B (local)	Dense page, oversized context	Unchanged	≈ 90 s
Groq Llama 4 (cloud)	Bilingual French/Arabic	Not measurable: vision access withdrawn by the provider (HTTP 404)	

The 8B result is explained by the memory constraint rather than by the model itself: on a 6 GB GPU it is partially offloaded to the CPU, which more than doubles the processing time without improving completeness.

Table 5.2: Comparison of the two local models on the target hardware

Criterion	**Qwen3-VL 4B**	**Qwen3-VL 8B**
Speed on the 6 GB GPU	Faster (59 s)	Slower (135 s)
Placement on the GPU	Fully resident ($\approx$ 3.3 GB)	Partially offloaded to the CPU
Completeness observed	10/10	9/10
Relative strengths	Fine print, checkbox states	Recovery of headers
Retained as default	Yes	No

The evaluation produced the following observations:

- The local VLM engine transcribed dense, multilingual and handwritten pages with high fidelity, preserving structure and Arabic right-to-left text.
- On the target hardware, the 4B model was both **faster and at least as complete** as the 8B model, and was retained as the default.
- The two-model verification successfully flagged genuine omissions (for example a dropped header and a small table annotation) for review.
- Targeted prompt improvements measurably recovered specific classes of missed content, such as form helper text and checkbox states.

5.6 Limitations and Future Work

The main limitations, and the corresponding directions for improvement, are:

- Very faint handwriting can still be missed by a 4B model; a larger model or the cloud engine helps when accuracy outweighs confidentiality.
- Recognition speed depends on the GPU; a more powerful machine would allow larger models at the same latency.
- Future work includes surfacing the two-model comparison directly in the web interface, adding a human-review workflow, and extending the evaluation to a larger annotated corpus.

Chapter 6

Skills Developed

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During this internship, I had the opportunity to acquire and consolidate a set of both technical and transferable skills, directly related to applied artificial intelligence and document intelligence. These skills constitute a major asset for my future professional career.

6.1 Technical Skills

On the technical side, the main skills acquired or strengthened are the following:

- **Vision-Language Models and OCR**: understanding how modern VLMs read documents, and how they compare with classical OCR; selecting and prompting a model for a faithful, structure-preserving transcription.
- **Local model serving**: deploying and running a VLM locally with *Ollama*, and tuning inference parameters (context window, resolution) to fit a constrained GPU while preserving quality.
- **Back-end development**: building a REST API with *FastAPI*, including JWT authentication and integration with a *MongoDB* database.
- **Front-end development**: building a single-page application with *React* and *Vite*, including file upload, engine selection and a searchable history.
- **Image processing**: implementing a preprocessing pipeline (grayscale, denoising, deskewing, contrast enhancement) with *OpenCV* and *Pillow*, and rasterizing PDFs with *PyMuPDF*.
- **Multilingual and Arabic text handling**: managing mixed French/English/Arabic content, Unicode normalization and right-to-left rendering.
- **Evaluation and quality assurance**: implementing CER/WER metrics and designing a two-model verification method to detect potential omissions.

6.2 Transferable Skills

This internship also allowed the development of transferable skills essential in a professional context:

- **Teamwork**: collaboration with the AI and Data team of EY Tunisia.
- **Professional communication**: writing clear technical documentation and presenting results and design trade-offs to stakeholders.
- **Problem solving**: diagnosing performance and quality issues through measurement rather than guesswork, and iterating on concrete evidence.
- **Time and priority management**: organizing the work across the full lifecycle of the project.

- **Adaptability**: quickly mastering new tools, frameworks and technical environments.
- **Analytical and synthesis skills**: understanding a complex problem and proposing pragmatic, defensible solutions.

General Conclusion

At the end of this internship within EY Tunisia, I was able to discover and understand the challenges of applied artificial intelligence in a demanding professional-services environment, where confidentiality and reliability are paramount. Building an OCR system on top of Vision-Language Models illustrated how recent AI can solve concrete document-processing problems that classical techniques handle poorly, particularly for degraded, structured and multilingual (French, English and Arabic) content.

The project I carried out – an intelligent, multilingual OCR platform whose recognition engine runs a Vision-Language Model **locally and offline** – made it possible to transcribe difficult scanned documents while keeping confidential material on the machine. Beyond the recognition core, I delivered a complete platform (React, FastAPI, MongoDB), tuned it for constrained hardware, and, importantly, built the means to **measure** its quality and **verify** its completeness rather than merely assert it. This experience gave me a concrete opportunity to put theoretical concepts into practice and a better understanding of the operational and technical trade-offs encountered on a daily basis.

In addition, this internship allowed me to develop specific technical skills – in Vision-Language Models, local model serving, full-stack development and evaluation – as well as transferable skills such as teamwork, communication and project management.

Looking ahead, this experience provides a solid foundation for continuing to grow in the field of applied AI and document intelligence, keeping in mind the importance of rigorous measurement, data confidentiality and continuous adaptation as models and tools evolve.

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Abstract

This report presents the work carried out during an internship within EY Tunisia (AMC Ernst & Young). The objective was to design and build an intelligent Optical Character Recognition (OCR) system able to transcribe scanned documents in French, English and Arabic – including mixed-language and right-to-left content – by leveraging modern Vision-Language Models (VLMs) rather than classical OCR alone. The delivered solution is a full-stack platform (React, FastAPI, MongoDB) whose OCR core runs a local VLM (Qwen3-VL served through Ollama) fully offline on the host machine, keeping confidential documents on-device, with an optional cloud engine (Groq) for benchmarking. An image-preprocessing pipeline, a per-page reading strategy for PDFs, and a two-model completeness-verification tool were developed and evaluated with Character and Word Error Rate metrics. The system reliably extracts structured, multilingual text – including tables and checkbox states – from degraded and handwritten scans.

Keywords: OCR, Vision-Language Models, Qwen3-VL, Ollama, Multilingual, Arabic, Document Intelligence, FastAPI, React